Pool Ball Detection and Recognition

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Abstract

This report documents the experimental development of a two-stage computer vision pipeline for localising and recognising pool balls. Stage 1 tested a green-channel boost with adaptive thresholding, fixed HSV and single-channel lightness thresholding, and finally a successful multi-channel CIELAB edge-fusion detector followed by circular detection and geometric constraints. Stage 2 tested K-Means colour clustering and finally adopted one-dimensional hue and saturation histograms with a reduced circular sampling mask and a 30/70 weighting scheme. The final pipeline achieves Stage 1 precision 0.886, recall 0.867 and F1-score 0.876, together with Stage 2 recognition accuracy 70.5% across the six test tables.

1 Introduction

The task was to localise every pool ball on a rectified table image and then assign each surviving detection to one of the sixteen known balls. The six test images contain strong shadows, uneven lighting, tightly packed racked balls, and a mixture of solids and stripes. I developed the final system through a sequence of explicit experiments. Each failed design is documented because its failure explains the final architecture: Stage 1 became a colour-boundary fusion problem, while Stage 2 became a saturation-distribution problem.

2 Stage 1: Ball Detection

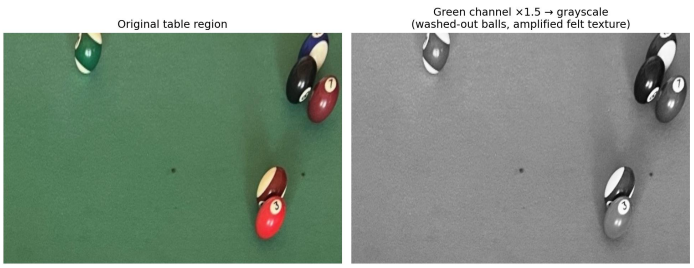
2.1 Perspective rectification

Each input photograph is first mapped by a planar homography from the four measured cushion-to-cushion corners into a 1434×683 metric table frame. The destination dimensions match the physical $1434\,\text{mm} \times 683\,\text{mm}$ playing area, so one transformed pixel corresponds to one millimetre. This metric frame is used by all later stages.

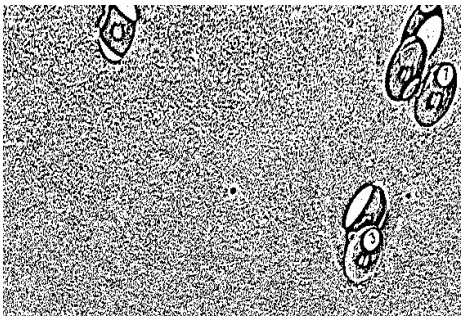
2.2 Failed green-channel boost and adaptive thresholding

To solve the problem of missing green balls, I initially hypothesised that isolating and boosting the RGB green channel by a factor of 1.5, followed by adaptive Gaussian thresholding, would force the balls to pop against the felt. The adaptive threshold used a

local block size of 15 and a constant of 2. This failed because boosting the green channel equally boosted the green table, and adaptive thresholding amplified the woven texture of the felt into severe noise. Figure 1 shows the original region and the green-boosted grayscale image, together with the resulting adaptive threshold map. The ball boundaries are washed out while the felt texture dominates the edge map.



(a) Original region versus green-channel boost.

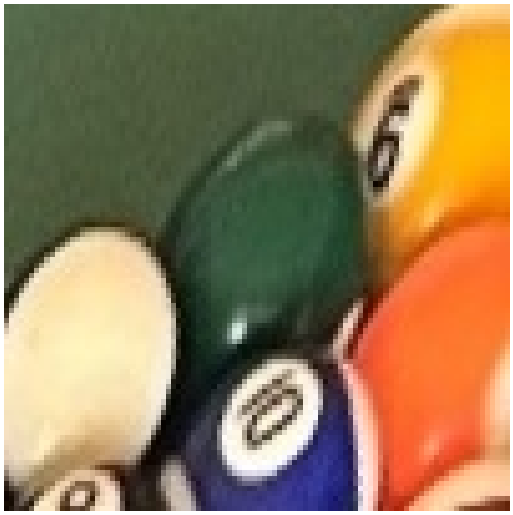


(b) Adaptive Gaussian threshold map: severe felt texture noise.

Figure 1: The green-channel boost and adaptive-threshold experiment. The boosted image washes out ball features, and the threshold map is dominated by felt texture rather than ball boundaries.

2.3 Failed HSV and single-channel lightness thresholding

I then attempted to isolate the felt directly in HSV space with a fixed hue range of 35–85, saturation range 40–255, and value range 40–255. This also failed for the green object balls, particularly balls 6 and 14, because their hue and saturation overlap almost completely with the green felt. Figure 2 shows a 100×100 region around the green ball and the resulting binary mask. In the mask, the green ball vanishes into the white felt instead of producing a distinct foreground blob. A single-channel lightness threshold produced the complementary failure: shadows fragmented the ball boundaries and the threshold could not separate dark balls from the table. Both results motivated a multi-channel approach.



(a) Original region around the green ball.



(b) Fixed HSV threshold: the green ball merges into the felt.

Figure 2: Visual evidence for the failure of fixed HSV thresholding. The green ball and the green felt are not separable with static hue and saturation limits.

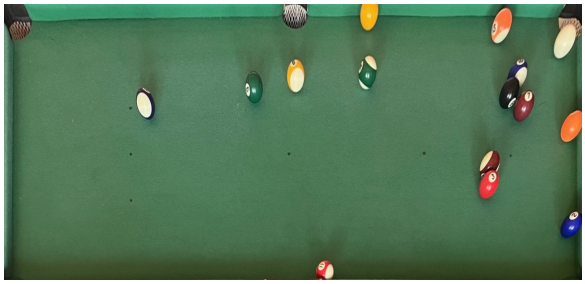
2.4 Successful multi-channel CIELAB edge fusion

I realised that the solution required looking at colour boundaries, not just brightness. I split the rectified image into the CIELAB colour space and applied Gaussian smoothing with a 5×5 kernel and a standard deviation of 1.2 to all three channels. I then ran Canny edge detection independently on each channel. I used thresholds of 45 and 95 for the lightness channel L, and more sensitive thresholds of 35 and 85 for the chromatic channels a and b. The three binary edge maps were combined with a logical OR to produce a highly robust gradient map that captured subtle colour shifts as well as luminance boundaries.

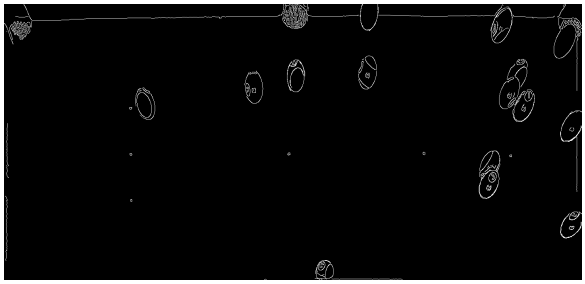
Key Idea: LAB Multi-Channel Edge Fusion

The lightness channel supplies the strong outer silhouette of each ball, while the chromatic channels supply the colour boundaries that remain visible when a ball is dark, shadowed, or similar in brightness to the felt. Fusing the three edge maps produces a gradient image that is markedly more complete than any single-channel edge map.

Figure 3 shows the rectified Table 4 image and the corresponding fused edge map used by the circular detector.



(a) Rectified Table 4.



(b) Multi-channel LAB edge fusion map.

Figure 3: Rectification and the successful fused edge map for Table 4.

2.5 Circular detection and geometric constraints

The fused edge map is passed to the Hough Circle Transform with a minimum centre separation of 40 pixels and a ball-radius range of 19 to 24 pixels. Two geometric constraints remove candidates that cannot correspond to balls resting on the playing surface. A pocket constraint rejects centres within 95 pixels of the six theoretical pocket positions. A cushion constraint rejects centres within 22 pixels of the outer border, because a ball frozen to the cushion has its centre approximately one radius from the cushion and the wooden rails generate non-felt edges.

2.6 Stage 1 results

Table 1 reports the final Stage 1 metrics. Overall precision is 0.886, recall is 0.867, and the F1-score is 0.876.

Table 1: Stage 1 ball-detection metrics after multi-channel LAB edge fusion.

Image	TP	FP	FN	Precision	Recall	F1-score
Table 1	16	1	0	0.941	1.000	0.970
Table 2	15	3	1	0.833	0.938	0.882
Table 3	15	0	1	1.000	0.938	0.968
Table 4	12	2	3	0.857	0.800	0.828
Table 5	10	2	4	0.833	0.714	0.769
Table 6	10	2	3	0.833	0.769	0.800
Overall	78	10	12	0.886	0.867	0.876

3 Stage 2: Ball Recognition and Classification

3.1 Failed K-Means clustering

To differentiate striped from solid balls, I first attempted to use K-Means clustering with $K = 2$ on the pixels inside each ball mask to extract two dominant colours, then

compared those colours with the reference balls using Euclidean distance. This proved computationally heavy and unreliable because the visibility of the white stripe depends on the physical rotation of the ball. Figure 4 shows the result on a solid blue ball. The two clusters split the number and the specular reflections rather than producing a stable stripe-versus-solid descriptor.

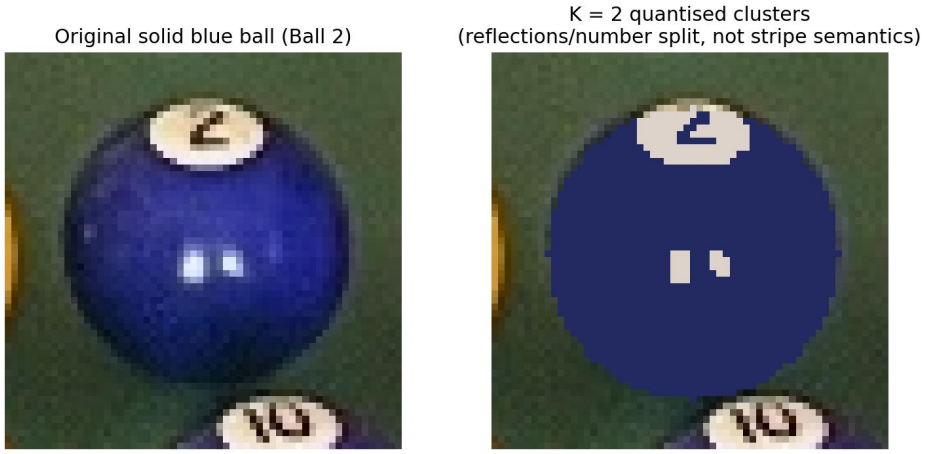


Figure 4: K-Means ($K = 2$) applied to a solid blue ball. The clustering splits the number and reflections, not a rotationally stable stripe/solid feature.

3.2 Successful saturation-dominant histograms

I pivoted to one-dimensional histograms. I extracted a hue histogram with 30 bins and a saturation histogram with 32 bins from a reduced circular sampling mask of radius 16 pixels. The reduced mask prevents a slightly off-centre detection from admitting felt, shadow, or corner pixels into the descriptor. I then compared test histograms with the sixteen reference histograms using the Bhattacharyya distance. Because a white stripe drastically lowers a ball's average saturation regardless of rotation, I applied a custom weighting system: 30% weight to hue, 70% weight to saturation, and 0% weight to brightness.

Saturation-Dominant Distance

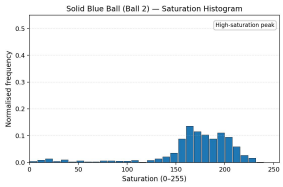
$$d = 0.30\, d_{\text{hue}} + 0.70\, d_{\text{sat}}. \tag{1}$$

A detection whose weighted distance exceeds 0.80 is rejected as a non-ball or as an unreliable observation.

Figure 5 compares the solid blue ball 2 with the striped blue ball 10. The solid ball peaks at high saturation, whereas the striped ball has a pronounced low-saturation peak caused by its white stripe. This shift is independent of the ball's physical rotation and provides a mathematically robust feature for distinguishing stripes from solids.



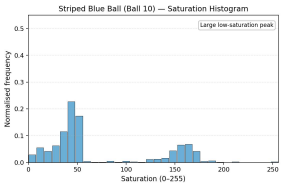
(a) Ball 2 crop.



(b) Solid histogram.



(c) Ball 10 crop.



(d) Striped histogram.

Figure 5: Solid versus striped blue balls and their saturation histograms. The white stripe shifts the striped ball's distribution strongly toward the low-saturation end, motivating the 70% saturation weighting.

3.3 Stage 2 results

Table 2 reports recognition accuracy for the true-positive detections. Across all six tables, 55 of 78 correctly localised balls receive the correct label, giving an overall recognition accuracy of 70.5%. Figure 6 shows the final labelled output for Table 4.

Table 2: Stage 2 recognition accuracy on correctly localised balls.

Image	Correct labels	True positives	Accuracy
Table 1	11	16	0.688
Table 2	9	15	0.600
Table 3	11	15	0.733
Table 4	10	12	0.833
Table 5	7	10	0.700
Table 6	7	10	0.700
Overall	55	78	0.705

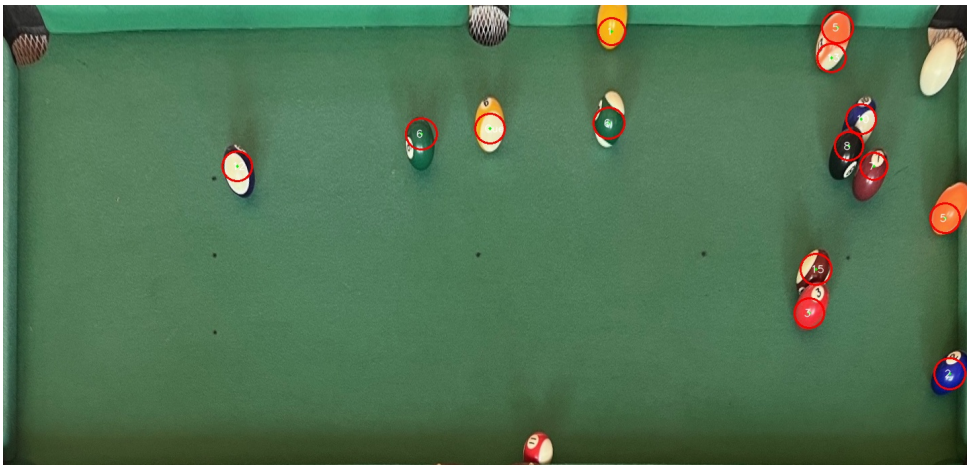


Figure 6: Final labelled output for Table 4 after Stage 2 saturation-dominant classification.

4 Discussion

The multi-channel LAB edge-fusion detector clearly outperformed the green-boost, adaptive-threshold, fixed-HSV, and single-channel lightness experiments. The lightness channel supplies the outer silhouette of each ball, while the a and b channels contribute colour boundaries that remain visible when a ball is dark, shadowed, or similar in brightness to the felt. The pocket and cushion constraints remove geometrically impossible responses, and the minimum centre separation removes duplicate detections. The resulting Stage 1 F1-score of 0.876 exceeds the previous single-channel baseline of approximately 0.831, and recall rises to 0.867.

Recognition is limited mainly by the ambiguity between similarly coloured solids and stripes. The saturation-dominant histogram improves on K-Means, but residual errors on Tables 1, 2, 5, and 6 show that a global hue and saturation distribution still cannot fully capture the spatial arrangement of the stripe and number. Higher-resolution descriptors that preserve spatial structure, or a dedicated stripe detector, would be the natural next step.

5 Conclusion

I replaced the failing HSV and brightness-threshold designs with a multi-channel CIELAB edge-fusion detector, and I replaced the unstable K-Means recognition with a 30/70 hue–saturation histogram descriptor. The final system achieves Stage 1 precision 0.886, recall 0.867, and F1 0.876, together with Stage 2 recognition accuracy 0.705. The experiment confirms that colour-boundary fusion is more robust than static colour thresholding for this dataset, and that saturation is a more reliable cue than hue for distinguishing striped from solid balls.